

CSA15 Cloud Computing and Big Data Analytics

Local IaaS Operations Lab: Snapshot, Clone and Restore Testing

Course Code	CSA15	Course Title	Cloud Computing and Big Data Analytics
Guided Project	GP-VM-03	Submission	Individual evidence through GitHub and Google Classroom
Faculty	Dr C. Rajesh Babu	Employee ID	SSETSCS415
Target Learner	Beginner CSE student	Estimated Duration	2 to 3 lab hours
Primary Tool	Oracle VirtualBox or VMware Workstation	Cloud Link	Builds operational recovery thinking before cloud VM backup and image management

Guided Project Pattern

Read the idea, perform the action, observe the result, record evidence, explain what happened, extend the setup to your capstone context, and submit the final proof through GitHub and Google Classroom.

Learning Outcomes

- Create a safe baseline snapshot of a configured VM.
- Test snapshot recovery by making a visible temporary change and restoring the previous state.
- Create or plan a full/linked clone according to storage availability and tool support.
- Compare snapshot, clone and VM image/export in operational terms.
- Prepare a complete VM recovery evidence portfolio.

Before You Start

- Completed VM from GP-VM-01 and configuration/storage evidence from GP-VM-02.
- At least 5 GB free storage for clone testing where possible; otherwise prepare a justified clone plan.
- The VM must not contain passwords, keys or personal sensitive data in screenshots.
- GitHub account and Google Classroom access.

Quick Learning Notes

- A snapshot records a point-in-time VM state and is useful before risky changes.
- A clone creates another VM copy for parallel testing or repeated experiments.
- A full clone is independent and consumes more storage. A linked clone depends on the base VM and usually consumes less storage.
- A restore test is important because a snapshot is useful only if it can actually return the VM to an earlier state.

Lab Exercise Coverage

Lab Exercise	How This Guided Project Covers It
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Experiment 10	Create a snapshot and test it by loading/restoring the previous version.
Experiment 11	Create a clone of a VM and test it by opening the cloned VM.

Hypervisor Choice

Tool	When to Choose It	Student Evidence
Oracle VirtualBox	Use Snapshots from the VM tools/menu and Clone from Machine options.	Snapshot list, restore test, clone window and cloned VM proof.
VMware Workstation	Use Snapshot Manager / Take Snapshot and VM -> Manage -> Clone where available.	Snapshot manager, revert proof, clone wizard and cloned VM proof.
Safety Rule Do not delete the original VM while testing snapshots or clones. If storage is low, record a clone plan with menu evidence instead of creating a large duplicate.		

Architecture View

Base VM	Baseline Snapshot	Temporary Change	Restore Test	Clone/Plan	Recovery Evidence
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The recovery workflow begins with a stable VM, captures a baseline, performs a controlled change, restores the previous version and then prepares a duplicate environment or clone plan.

Guided Build

Stage 1 - Prepare the Baseline VM

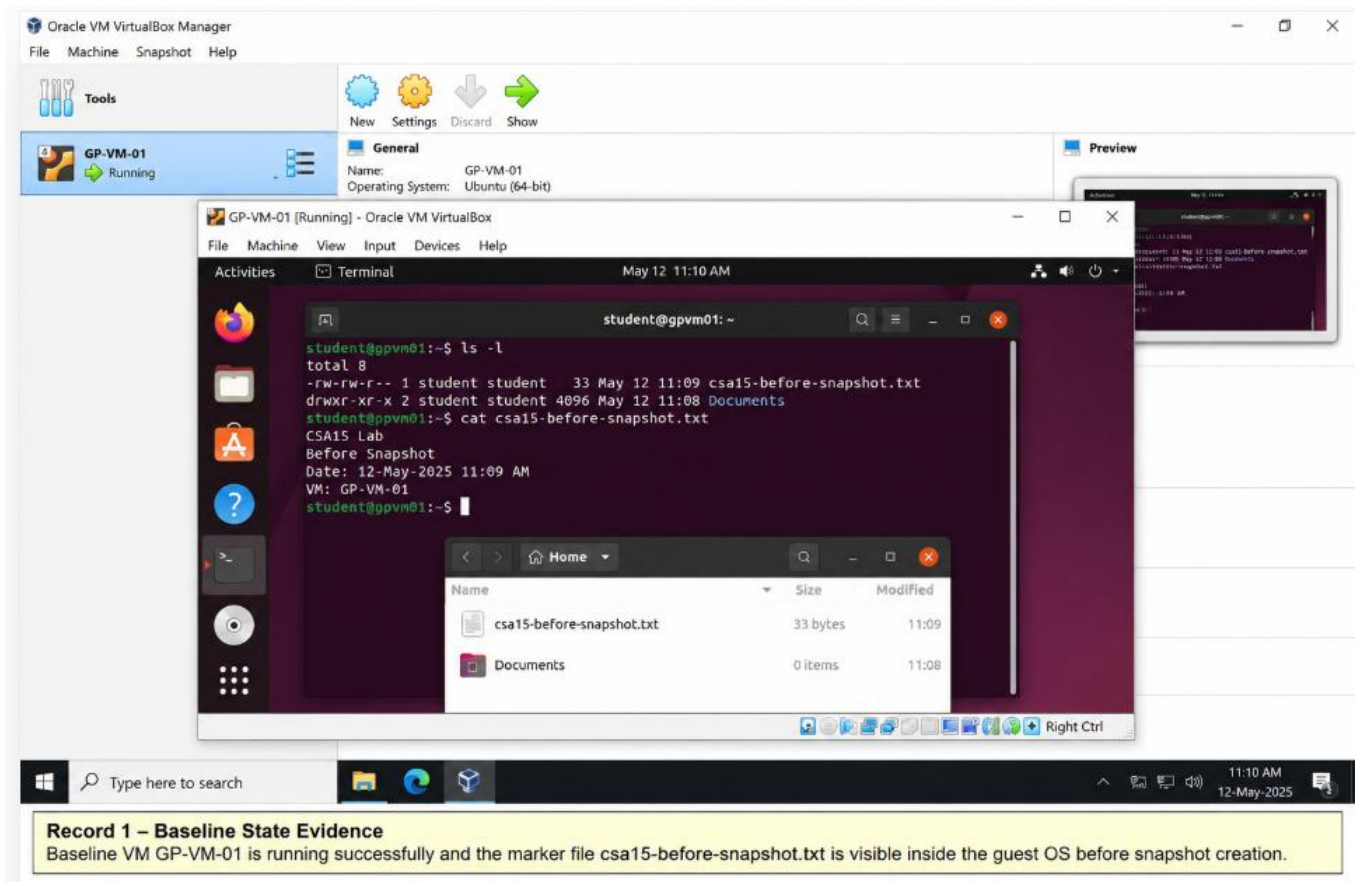
Learn: Snapshot and clone testing must begin from a known working VM state.

- Open the VM in VirtualBox or VMware.
- Boot the guest OS and confirm that it reaches desktop, shell or login prompt.
- Create a visible marker file inside the guest OS if possible, such as csa15-before-snapshot.txt.
- Shut down or pause the VM according to the hypervisor recommendation.

Checkpoint

A stable baseline VM is ready and the before-snapshot state is visible.

Record 1 - Baseline state evidence



Stage 2 - Take a Snapshot

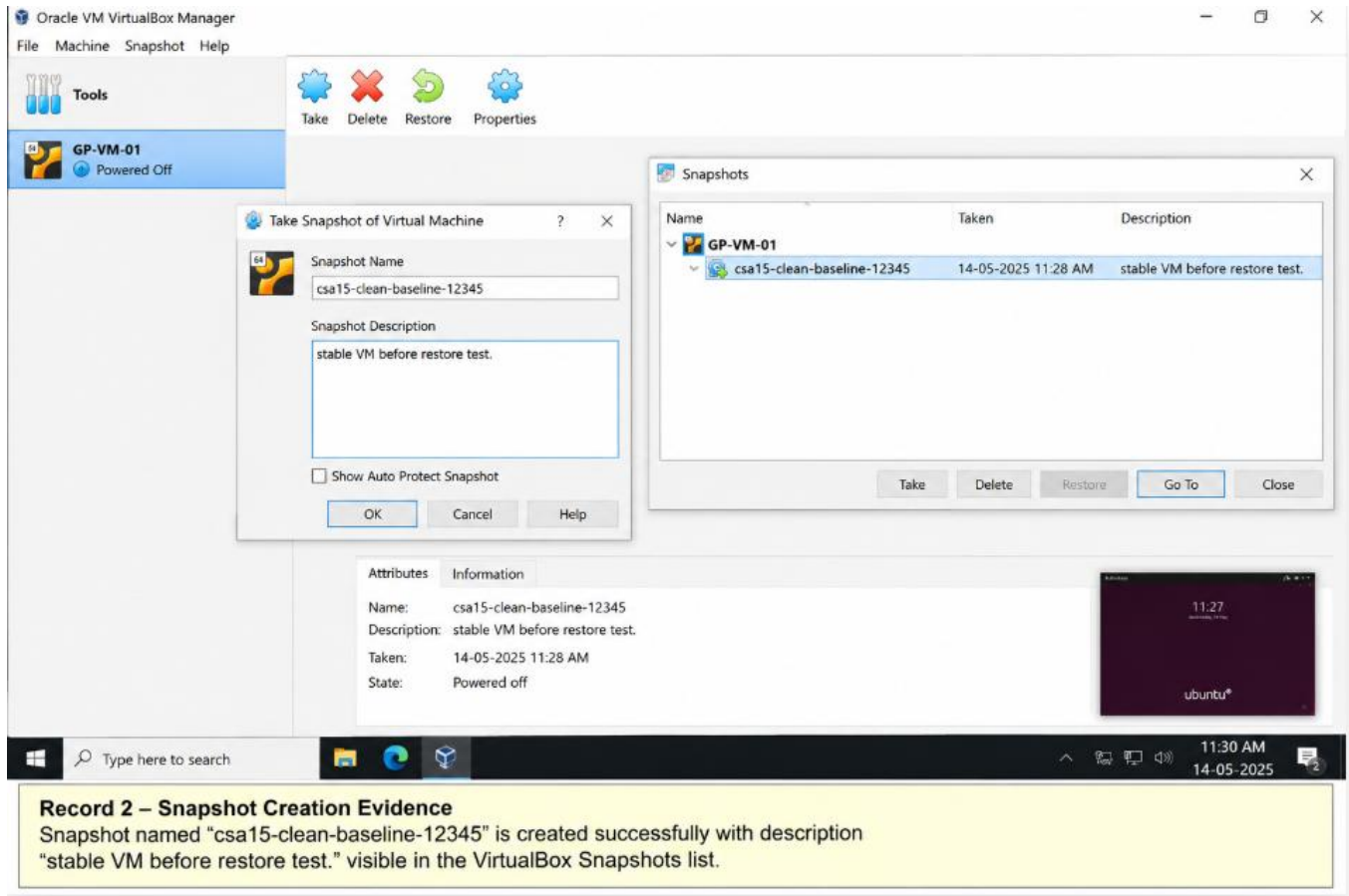
Learn: A snapshot preserves the current VM state before a risky action.

- In VirtualBox, select the VM and open Snapshots, then choose Take.
- In VMware, select VM -> Snapshot -> Take Snapshot, or open Snapshot Manager.
- Name the snapshot csa15-clean-baseline-register-number.
- Write a short description: stable VM before restore test.
- Capture the snapshot list or manager screen.

Checkpoint

Three named baseline snapshot is visible in the hypervisor.

Record 2 - Snapshot creation evidence



Record 2 – Snapshot Creation Evidence
 Snapshot named "csa15-clean-baseline-12345" is created successfully with description "stable VM before restore test." visible in the VirtualBox Snapshots list.

Stage 3 - Make a Controlled Change and Restore

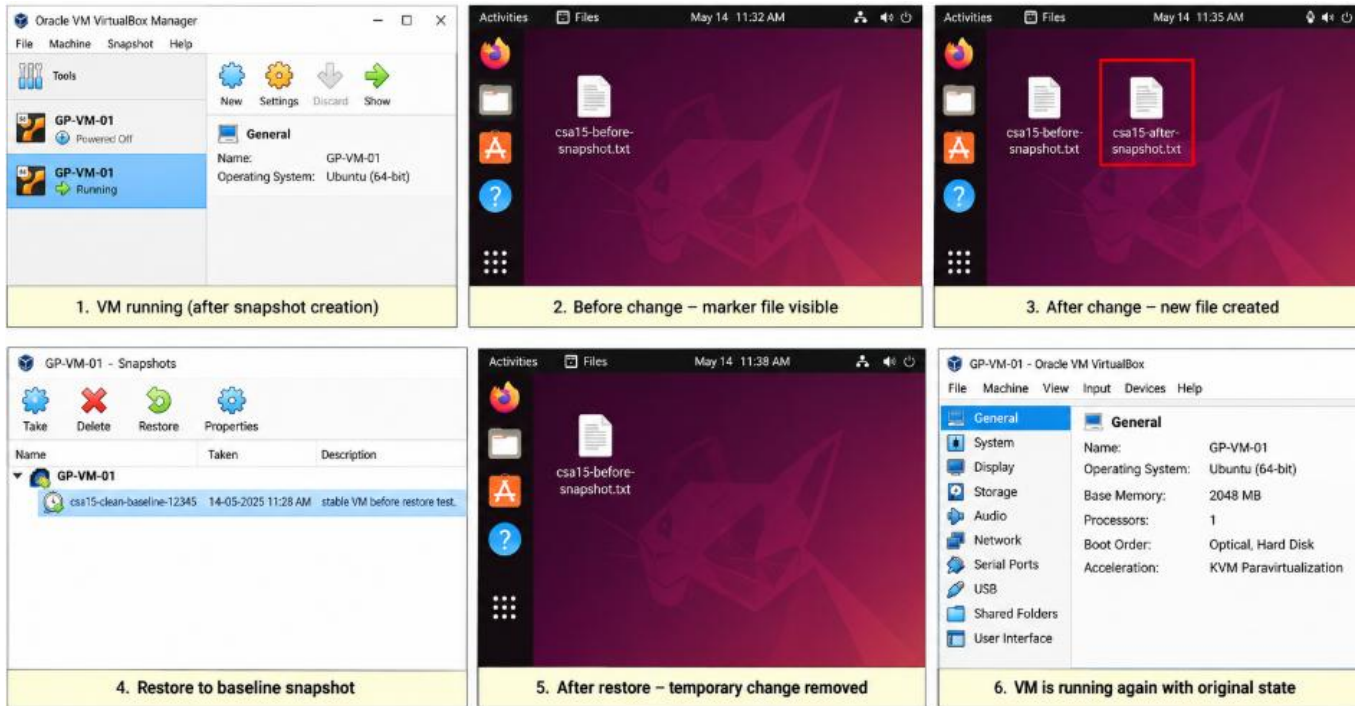
Learn: A restore test proves that the snapshot is usable.

- Boot the VM after snapshot creation.
- Create a temporary visible change such as csa15-after-snapshot.txt, a desktop folder or a small configuration note.
- Capture evidence of the temporary change.
- Shut down or pause the VM and restore/revert to the baseline snapshot.
- Boot again and verify whether the temporary change disappeared or the previous state returned.

Checkpoint

The restore result is documented with before-change, after-change and after-restore evidence.

Record 3 - Snapshot restore test evidence

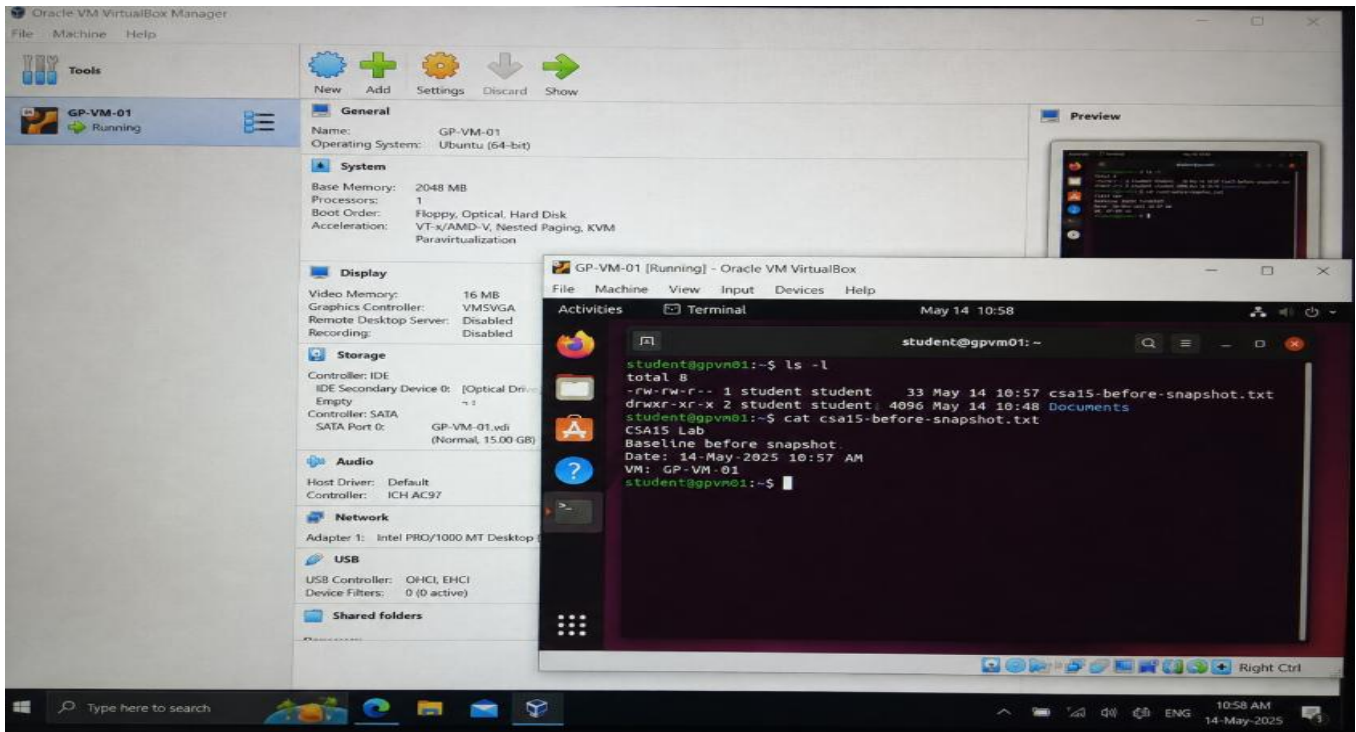


Stage 4 - Create or Plan a Clone

Learn: A clone creates a second VM environment that can be used without disturbing the base VM.

- In VirtualBox, choose Clone from Machine tools/menu and name it CSA15-Clone-RegisterNumber.
- In VMware, choose VM -> Manage -> Clone if the installed edition supports it.
- Choose full clone if sufficient storage is available. Choose linked clone only when the tool supports it and the base VM will remain available.
- If the tool edition or storage does not allow cloning, record the exact limitation and write a clone plan.
- Open the cloned VM or show the completed clone/plan evidence.

Record 4 - Clone evidence or clone plan



Stage 5 - Build the Recovery Portfolio

Learn: Professional VM operations require evidence, decision reasoning and cleanup status.

- Prepare a comparison table: snapshot, full clone, linked clone, exported image and rebuild from ISO.
- Record storage impact: original VM size, snapshot impact and clone/export estimate where visible.
- Write when you would use snapshot, clone or rebuild for your capstone project.
- Remove only temporary cloned/exported files created for this lab if storage is low and after evidence is captured.

Checkpoint

Three portfolio proves snapshot creation, restore testing, clone/plan and capstone recovery reasoning.

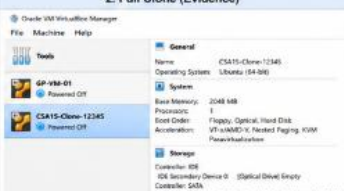
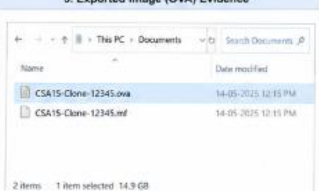
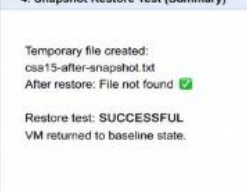
Record 5 - Final recovery portfolio

Record 5 – Final Recovery Portfolio

A. RECOVERY METHODS COMPARISON						
Method	Description	Storage Impact	Speed (Recovery)	When to Use	Advantages	Limitations
Snapshot	Captures the current state of the VM (disk + memory + device state).	Low (differential) Uses only changed data blocks.	Very Fast (Seconds)	Before risky changes, software updates, configuration experiments.	Quick, easy, minimal storage, ideal for short-term recovery.	Tied to the base VM. Multiple snapshots consume more space over time.
Full Clone	Creates a completely independent copy of the VM with full data.	High (Almost same as original VM size)	Medium (Minutes)	Testing, development, parallel environments, compatibility tests.	Independent, safe to modify, original VM not affected.	Consumes significant storage and time to create.
Linked Clone	Shares disk blocks with the original VM. Changes stored in a separate file.	Low to Medium (Depends on changes)	Medium (Minutes)	When storage is limited but VM stays available and stable.	Faster than full clone, saves storage initially.	Depends on the base VM. If base VM is removed/corrupted, clone breaks.
Exported Image (OVA/OVF)	Exports VM as a single portable file.	High (Similar to full VM size)	Slow (Minutes)	Backup, migration, sharing VM to another host.	Portable, can be moved to different systems.	Large file size, export/import takes time.
Rebuild from ISO	Fresh installation using original ISO and configuration steps.	Low (Only new VM disk size)	Slow (Hours)	When VM is severely corrupted or backup/clone not available.	Clean environment, no dependencies.	Time consuming, need to reinstall OS and applications.

B. STORAGE IMPACT SUMMARY			
Item	Details	Size (Approx.)	Notes / Evidence
Original VM (GP-VM-01)	VDI Disk (Normal)	15.00 GB	From VirtualBox Storage Settings
Snapshot (csa15-clean-baseline-12345)	Differential	210 MB	After snapshot creation
Full Clone (CSA15-Clone-12345)	VDI Disk (Normal)	14.92 GB	Independent full clone
Exported Image (OVA)	OVA File	14.95 GB	Exported to Documents folder

C. WHEN TO USE WHAT (For Capstone Project)	
Snapshot: Before major configuration changes, software updates, or risky testing in MediQueue AI backend. Clone: Create a clone for parallel development and testing of new features without affecting the stable base VM. Exported Image: Keep periodic exports as backups for migration or restoration on another machine/cloud. Rebuild from ISO: Use only when the VM is corrupted or when a completely clean environment is required for deployment.	

D. EVIDENCE GALLERY			
1. Snapshot List (Evidence)	2. Full Clone (Evidence)	3. Exported Image (OVA) Evidence	4. Snapshot Restore Test (Summary)
			

E. CLEANUP STATUS				
Temporary files created for this lab:				
Item	Purpose	Status	Action Taken	Notes
CSA15-Clone-12345 (Full Clone)	Testing clone functionality	Deleted	VM deleted from VirtualBox	Space was low after testing.
CSA15-Clone-12345.ova (Exported Image)	Testing export/import	Deleted	OVA file removed from Documents	Evidence captured before deletion.
csa15-after-snapshot.txt (Temporary File)	Snapshot restore test	N/A (Already removed by restore)	Not applicable	File disappeared after restore (expected).
Checkpoint: Snapshot creation, restore test, clone/plan and recovery reasoning are documented. Temporary data cleaned up after capturing evidence.				

Final Evidence Checklist

Done	Checkpoint	Evidence to Capture
☐	Baseline prepared	Stable VM and before-snapshot marker evidence.
☐	Snapshot created	Snapshot manager/list screenshot with correct name.
☐	Restore tested	Before-change, after-change and after-restore evidence.
☐	Clone completed/planned	Clone proof or justified clone limitation/plan.
☐	Recovery reasoning written	Snapshot vs clone vs image/export comparison and capstone link.

Explain and Extend

Explain why a snapshot is not the same as a full backup.

A **snapshot** saves the VM's current state for quick rollback, while a **full backup** creates an independent copy of the VM/data.

Snapshot = quick recovery

Full backup = independent protection

Explain when a full clone is better than a linked clone.

A **full clone** is better when you need an **independent VM** that can work without the original VM.

Example: Use a full clone for testing major changes because deleting or changing the original VM will not affect the clone.

Explain how restore testing reduces risk during capstone development.

Restore testing reduces risk by confirming that a snapshot or backup can actually return the VM to a working state after a failed change.

Example: If a software update breaks the capstone VM, the tested snapshot can restore the previous working state quickly.

Extend this setup to your capstone: what change would you test only after taking a snapshot?

I would take a snapshot before making **major software updates, configuration changes, or installing new applications** in the capstone VM.

Example: Before changing the database or server configuration, create a snapshot so the VM can be restored if something goes wrong.

Submission Protocol

- Create a GitHub folder named CSA15-GP-VM03-Snapshot-Clone-Restore-Test.
- Upload this completed document as DOCX/PDF along with screenshots and short README.md.
- README.md must include tool used, VM/cloud specifications, configuration decisions, screenshots index and cleanup status.
- Submit the GitHub repository link in Google Classroom. Do not upload passwords, private keys, tokens, public IP credentials or sensitive account screenshots.

Student Assurance

Declaration

I confirm that this guided-project evidence was completed by me, the screenshots are from my own lab environment, and no secret keys, passwords or sensitive account information are included.

Student Signature: **T.PRAKEERTHI** Register Number:**192524279** Date: **14/08/2026**